

一、[参考解析] (1) 因为 $\left(1 + \frac{1}{n}\right)^\alpha < 1 + \frac{1}{n}$, 所以

$$0 < (n+1)^\alpha - n^\alpha = n^\alpha \left(\left(1 + \frac{1}{n}\right)^\alpha - 1 \right) < n^\alpha \left(\left(1 + \frac{1}{n}\right) - 1 \right) = \frac{1}{n^{1-\alpha}}.$$

显然 $\lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} = 0$, 故由夹逼准则可知

$$\lim_{n \rightarrow \infty} ((n+1)^\alpha - n^\alpha) = 0.$$

(2) 当 $t = 0$ 时, $x = 1, y = 0$. 对 $x = t + \cos t$ 的两边关于 t 求导, 得

$$\frac{dx}{dt} = 1 - \sin t, \quad \left. \frac{dx}{dt} \right|_{t=0} = 1.$$

再对 $e^y + ty + \sin t = 1$ 的两边关于 t 求导, 得

$$e^y \frac{dy}{dt} + y + t \frac{dy}{dt} + \cos t = 0, \quad \left. \frac{dy}{dt} \right|_{t=0} = -1,$$

从而 $\left. \frac{dy}{dx} \right|_{t=0} = -1$, 所以切线方程为 $y - 0 = -(x - 1)$, 即 $x + y = 1$.

(3) [方法一]

$$\begin{aligned} I &= \int \frac{\ln(x + \sqrt{1+x^2})}{(1+x^2)^{3/2}} dx \stackrel{x=\tan t}{=} \int \frac{\ln(\tan t + \sec t)}{\sec t} dt \\ &= \int \ln(\tan t + \sec t) d(\sin t) \\ &= \sin t \ln(\tan t + \sec t) - \int \sin t d[\ln(\tan t + \sec t)] \\ &= \sin t \ln(\tan t + \sec t) - \int \sin t \frac{1}{\tan t + \sec t} (\sec^2 t + \tan t \sec t) dt \\ &= \sin t \ln(\tan t + \sec t) - \int \frac{\sin t}{\cos t} dt \\ &= \sin t \ln(\tan t + \sec t) + \ln |\cos t| + C \\ &= \frac{x}{\sqrt{1+x^2}} \ln(x + \sqrt{1+x^2}) - \frac{1}{2} \ln(1+x^2) + C. \end{aligned}$$

[方法二]

$$\begin{aligned} I &= \int \frac{\ln(x + \sqrt{1+x^2})}{(1+x^2)^{3/2}} dx = \int \ln(x + \sqrt{1+x^2}) d\frac{x}{\sqrt{1+x^2}} \\ &= \frac{x}{\sqrt{1+x^2}} \ln(x + \sqrt{1+x^2}) - \int \frac{x}{\sqrt{1+x^2}} - \frac{1}{x + \sqrt{1+x^2}} \left(1 + \frac{x}{\sqrt{1+x^2}}\right) dx \\ &= \frac{x}{\sqrt{1+x^2}} \ln(x + \sqrt{1+x^2}) - \int \frac{x}{1+x^2} dx \\ &= \frac{x}{\sqrt{1+x^2}} \ln(x + \sqrt{1+x^2}) - \frac{1}{2} \ln(1+x^2) + C. \end{aligned}$$

(4)

$$\begin{aligned} &\lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x} \sqrt[3]{\cos 3x}}{x^2} \\ &= \lim_{x \rightarrow 0} \left[\frac{1 - \cos x}{x^2} + \frac{\cos x (1 - \sqrt{\cos 2x} \sqrt[3]{\cos 3x})}{x^2} \right] \\ &= \frac{1}{2} + \lim_{x \rightarrow 0} \left[\frac{\cos x (1 - \sqrt{\cos 2x} \sqrt[3]{\cos 3x})}{x^2} \right] \\ &= \frac{1}{2} + \lim_{x \rightarrow 0} \left[\frac{1 - \sqrt{(\cos 2x - 1) + 1}}{x^2} + \frac{1 - \sqrt[3]{(\cos 3x - 1) + 1}}{x^2} \right] \\ &= \frac{1}{2} + \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{2x^2} + \lim_{x \rightarrow 0} \frac{1 - \cos 3x}{3x^2} = \frac{1}{2} + 1 + \frac{3}{2} = 3. \end{aligned}$$

二、[参考解析] 设

$$P(x, y) = y(2 - f(x^2 - y^2)), \quad Q(x, y) = xf(x^2 - y^2).$$

由题设可知, 积分与路径无关, 于是有 $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$, 由此可知

$$(x^2 - y^2)f'(x^2 - y^2) + f(x^2 - y^2) = 1.$$

记 $t = x^2 - y^2$, 则得微分方程 $tf'(t) + f(t) = 1$. 凑微分得 $(tf(t))' = 1$, 所以

$$tf(t) = t + C.$$

由 $f(1) = 0$, 得 $C = -1$, 所以 $f(t) = 1 - \frac{1}{t}$. 从而

$$f(x^2 - y^2) = 1 - \frac{1}{x^2 - y^2}.$$

三、[参考证明] 由 Cauchy 不等式可得

$$\int_0^1 f(x)dx \int_0^1 \frac{1}{f(x)}dx \geq \left(\int_0^1 \sqrt{f(x)} \sqrt{\frac{1}{f(x)}} dx \right)^2 = 1.$$

又由于

$$(f(x) - 1)(f(x) - 3) \leq 0,$$

则

$$(f(x) - 1)(f(x) - 3)/f(x) \leq 0,$$

即

$$f(x) + \frac{3}{f(x)} \leq 4, \quad \text{且} \int_0^1 \left(f(x) + \frac{3}{f(x)} \right) dx \leq 4.$$

由于

$$\int_0^1 f(x)dx \int_0^1 \frac{1}{f(x)}dx \leq \frac{1}{3} \left(\frac{1}{2} \int_0^1 \left(f(x) + \frac{3}{f(x)} \right) dx \right)^2,$$

故

$$1 \leq \int_0^1 f(x)dx \int_0^1 \frac{1}{f(x)}dx \leq \frac{4}{3}.$$

四、[参考解析] 为了便于计算, 将区域 V 割补成三部分.

第一部分: 整个大球 $x^2 + y^2 + (z - 1)^2 \leq 9$, 记为 V_1 , 用广义球坐标表示, 即

$$\begin{cases} x = r \cos \theta \sin \varphi, & y = r \sin \theta \sin \varphi, & z - 1 = r \cos \varphi, \\ 0 \leq \theta \leq 2\pi, & 0 \leq \varphi \leq \pi, & 0 \leq r \leq 3. \end{cases}$$

$$I_1 = \iiint_{V_1} (x^2 + y^2) dV = \int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^3 r^2 \sin^2 \varphi r^2 \sin \varphi dr = \frac{648\pi}{5};$$

第二部分: 小球 $x^2 + y^2 + (z - 2)^2 \leq 4$, 记为 V_2 , 用广义球坐标表示, 即

$$\begin{cases} x = r \cos \theta \sin \varphi, & y = r \sin \theta \sin \varphi, & z - 2 = r \cos \varphi, \\ 0 \leq \theta \leq 2\pi, & 0 \leq \varphi \leq \pi, & 0 \leq r \leq 2. \end{cases}$$

$$I_2 = \iiint_{V_2} (x^2 + y^2) dV = \int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^2 r^2 \sin^2 \varphi r^2 \sin \varphi dr = \frac{256\pi}{15};$$

第三部分: 大球 V_1 位于 xOy 平面以下的部分, 记为 V_3 , 用“先二后一”法计算.

$$I_3 = \iiint_{V_3} (x^2 + y^2) dV = \int_{-2}^0 dz \iint_{x^2 + y^2 \leq 9 - (z-1)^2} (x^2 + y^2) d\sigma$$

$$= \int_{-2}^0 dz \int_0^{2\pi} d\theta \int_0^{\sqrt{9-(z-1)^2}} r^3 dr = \frac{\pi}{2} \int_{-2}^0 (9 - (z-1)^2)^2 dz = \frac{136\pi}{5}.$$

综合上述, 得

$$\iiint_V (x^2 + y^2) dV = I_1 - I_2 - I_3 = \left(\frac{648}{5} - \frac{256}{15} - \frac{136}{5} \right) \pi = \frac{256}{3} \pi.$$

五、[参考证明] 作辅助函数

$$\varphi(t) = f(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1)),$$

显然 $\varphi(t)$ 在 $[0,1]$ 上可导. 根据 Lagrange 中值定理, 存在 $c \in (0,1)$, 使得

$$\begin{aligned} \varphi(1) - \varphi(0) &= \varphi'(c) = \frac{\partial f(u,v)}{\partial u} (x_2 - x_1) + \frac{\partial f(u,v)}{\partial v} (y_2 - y_1), \\ |\phi(1) - \phi(0)| &= |f(x_2, y_2) - f(x_1, y_1)| = \left| \frac{\partial f(u,v)}{\partial u} (x_2 - x_1) + \frac{\partial f(u,v)}{\partial v} (y_2 - y_1) \right| \\ &\leq \left(\left(\frac{\partial f(u,v)}{\partial u} \right)^2 + \left(\frac{\partial f(u,v)}{\partial v} \right)^2 \right)^{1/2} \left((x_2 - x_1)^2 + (y_2 - y_1)^2 \right)^{1/2} \\ &\leq M|AB|. \end{aligned}$$

六、[参考证明] 由于 $f(x)$ 在 $[0,1]$ 上连续, 所以积分 $\int_0^1 f(x)dx$ 存在. 根据定积分的定义, 得

$$\int_0^1 f(x)dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(x_k),$$

其中 $x_k \in \left[\frac{k-1}{n}, \frac{k}{n} \right]$. 利用平均值不等式

$$(f(x_1)f(x_2) \cdots f(x_n))^{\frac{1}{n}} \leq \frac{1}{n} \sum_{k=1}^n f(x_k)$$

及函数 $\ln x$ 的单调性, 得

$$\frac{1}{n} \sum_{k=1}^n \ln f(x_k) \leq \ln \left(\frac{1}{n} \sum_{k=1}^n f(x_k) \right).$$

两边取极限, 并利用函数 $\ln x$ 的连续性, 得

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln f(x_k) \leq \lim_{n \rightarrow \infty} \ln \left(\frac{1}{n} \sum_{k=1}^n f(x_k) \right) = \ln \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(x_k) \right),$$

仍根据定积分的定义, 上述不等式即

$$\int_0^1 \ln f(x) dx \leq \ln \int_0^1 f(x) dx.$$

七、[参考证明] 令 $S_0 = 0$, $S_k = \sum_{i=1}^k a_i b_i$, 则

$$a_k b_k = S_k - S_{k-1}, \quad a_k = \frac{S_k - S_{k-1}}{b_k}, \quad k = 1, 2, \dots.$$

因为

$$\sum_{k=1}^N a_k = \sum_{k=1}^N \frac{S_k - S_{k-1}}{b_k} = \sum_{k=1}^{N-1} \left(\frac{S_k}{b_k} - \frac{S_k}{b_{k+1}} \right) + \frac{S_N}{b_N} \geq \sum_{k=1}^{N-1} \frac{\delta}{b_k b_{k+1}} S_k,$$

所以 $\sum_{k=1}^{\infty} \frac{S_k}{b_k b_{k+1}}$ 收敛.

根据平均值不等式

$$\sqrt[k]{(a_1 a_2 \cdots a_k)(b_1 b_2 \cdots b_k)} \leq \frac{a_1 b_1 + a_2 b_2 + \cdots + a_k b_k}{k} = \frac{S_k}{k}$$

可得

$$\sum_{k=1}^{\infty} \frac{k \sqrt[k]{(a_1 a_2 \cdots a_k)(b_1 b_2 \cdots b_k)}}{b_{k+1} b_k} \leq \sum_{k=1}^{\infty} \frac{S_k}{b_{k+1} b_k},$$

故结论成立.